

Project 4.1

Superstore Sales Analytics Dashboard

Build a live, interactive retail analytics dashboard from raw data to deployment

10 Days · 3 Phases · Beginner Level · [Python](#) · [Streamlit](#) · [Pandas](#) · [NumPy](#) · [Matplotlib](#) · [Plotly](#)



10 Days

Duration



3 Phases

Structure



10,194 Rows

Dataset size



Streamlit Cloud

Final deployment

1. Project Overview

Project 4.1 takes you from a raw retail CSV file to a fully deployed, interactive web dashboard — entirely in Python. You will work through the complete data analytics pipeline: understanding the data, cleaning and enriching it, building charts, and publishing the result online for anyone to access.

Each of the 10 days adds one new capability to your application. By Day 10 you will have a multi-tab dashboard with live filters, Matplotlib and Plotly charts, NumPy-powered data quality alerts, and a public URL you can share in your portfolio.

What makes this project different

Most beginner projects stop at writing Python scripts. This project goes further — you build something that runs in a real browser, responds to user input, and can be shared with a public link. Every skill you practise here directly maps to how professional data analysts and BI developers work.

2. The Dataset

Field	Details
File name	samplesuperstore.csv (provided on Day 1)
Rows	~10,194 order lines
Columns	21 columns per row
Time period	January 2020 – December 2023
Geography	4 US regions — East, West, Central, South

Categories	3 product categories · 17 sub-categories
Key metrics	Sales (USD) · Profit (USD) · Discount (0–0.8) · Quantity
Date columns	Order Date · Ship Date (used for shipping-time analysis)

3. Project Structure — 3 Phases

The project is split into three phases. Each phase builds on the previous one. Do not skip ahead — the skills from Phase 1 are the building blocks for Phase 3.

PHASE 1 · Days 1 – 4

Streamlit Foundations

Learn every Streamlit widget through focused daily practice and guided mini-projects.

- Day 1 — Text and display elements (title, header, markdown, code, LaTeX)
- Day 2 — DataFrames, tables, metric KPI cards, and JSON display
- Day 3 — Input widgets: sliders, dropdowns, radio buttons, checkboxes
- Day 4 — Dates, file upload, download buttons, and colour picker

PHASE 2 · Day 5

Data & Preprocessing

Clean and enrich the raw CSV. The output of this phase feeds every Phase 3 chart.

- Load the raw CSV and run a full data health inspection
- Remove duplicates, fix data types, rename columns for cleaner code
- Engineer three new columns: Shipping Days, Order Year, Profit Margin %
- Export superstore_clean.csv — this file is used by every Day 6–10 script

PHASE 3 · Days 6 – 10

Dashboard Build

Add one new dashboard layer per day — layout, charts, rich visualisations, alerts, and live deployment.

- Day 6 — Sidebar filters, tab layout, live KPI cards
- Day 7 — Built-in Streamlit charts (line, bar, area)
- Day 8 — Professional charts with Matplotlib and Plotly Express
- Day 9 — NumPy data quality analysis and colour-coded alert banners
- Day 10 — Caching, download export, and deployment to Streamlit Cloud

4. Day-by-Day Plan

Day	Topic	You will learn	Daily deliverable
1	Text & Display	st.title, header, markdown, code, LaTeX	<i>Personal Profile Card</i>
2	Data Display	DataFrames, tables, metrics, JSON	<i>Student Grade Dashboard</i>
3	Input Widgets	Sliders, dropdowns, radio, checkbox	<i>BMI & Fitness Calculator</i>
4	Files & Dates	Upload, download, date picker, colour	<i>Expense Tracker App</i>
5	Preprocessing	Pandas clean, engineer, export CSV	<i>superstore_clean.csv</i>
6	Layout	Sidebar, tabs, columns, live filters	<i>Dashboard skeleton</i>
7	Built-in Charts	st.line_chart, bar_chart, area_chart	<i>Charts dashboard v1</i>
8	Rich Charts	Matplotlib bars, Plotly scatter, pie	<i>Charts dashboard v2</i>
9	Alerts	NumPy percentiles, z-scores, banners	<i>Dashboard with alerts</i>
10	Deployment	Caching, download, Streamlit Cloud	<i>Live dashboard URL</i> 🚀

5. Tools & One-Time Setup

Software you need

Python 3.10 or newer	Download from python.org — tick "Add to PATH" during installation
VS Code	Download from code.visualstudio.com — install the Python extension
Git for Windows	Download from git-scm.com/download/win — needed for Day 10 deployment
GitHub account	Sign up free at github.com — needed for Streamlit Cloud deployment

Python packages — install once

Open Command Prompt and run:

```
pip install streamlit pandas numpy matplotlib plotly
```

Folder structure for your project

```
superstore_app/
├── data/
│   ├── samplesuperstore.csv    ← provided on Day 1
│   └── superstore_clean.csv    ← you create this on Day 5
├── scripts/
│   └── prep.py                 ← preprocessing script (Day 5)
├── day01.py  day02.py  day03.py  day04.py
└── day05.py
```

```
|— day06.py day07.py day08.py day09.py
|— dashboard.py           ← final capstone (Day 10)
|— requirements.txt       ← needed for deployment
```

6. What You Will Be Able to Do

After completing Project 4.1 you will be able to:

Build & run	Create and launch a multi-page Streamlit web application from a Python script
Clean data	Load a raw CSV, identify and fix data quality issues, and engineer useful new columns using Pandas
Visualise	Build interactive charts using Streamlit built-ins, Matplotlib, and Plotly Express
Apply NumPy	Use NumPy for statistical analysis — percentiles, z-scores, and outlier detection
Design UX	Organise a dashboard with sidebar filters, tabs, column layouts, and KPI metric cards
Deploy	Push a project to GitHub and deploy a live public web app on Streamlit Cloud
Work professionally	Use consistent file naming, folder structure, and version control like a working analyst